

DATA STRUCTURES LAB

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS06	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

1. Introduce fundamental data structures, teach Stack and Queue ADTs, and explore Priority Queues.
2. Teach Linked List ADT, including various types, and demonstrate Stacks and Queues using linked lists.
3. Cover searching (linear and binary search) and sorting algorithms (bubble sort, quick sort, etc.), and compare their efficiency.
4. Explain tree concepts, binary trees, and various search trees (BST, AVL, etc.), and compare their properties.
5. Introduce graphs, their traversals, algorithms (Dijkstra's, Bellman Ford, etc.), and text processing (pattern matching, tries).

COURSE OUTCOMES

1. Students will implement Stacks, solve delimiter and postfix expression problems, implement Queues, and understand Priority Queues.
2. Students will implement linked lists and use them to create effective Stack and Queue data structures.
3. Students will apply search and sorting algorithms and make informed choices for specific problems.
4. Students will create, manipulate, and choose appropriate tree structures for different scenarios.
5. Students will work with graphs, perform traversals, apply graph algorithms, and implement text processing techniques.

LIST OF EXPERIMENTS

WEEK -1	INTRODUCTION TO STRUCTURES
	1. Write a program that implement structures (its operations) using i) Structures ii) Array of Structures iii) Nested Structures iv) Pointer to Structures
WEEK -2	STACK OPERATIONS
	2. Write a program that implement Stack (its operations) using array and linked list
WEEK -3	QUEUE OPERATIONS
	3. Write a program that implement Queue (its operations) using array and linked list
WEEK -4	SINGLY LINKED LIST DOUBLY LINKEDLIST
	4. Write a program that uses functions to perform the following operations on singly linkedlist & doubly linked list i) Creation ii) Insertion iii) Deletion iv) Traversal
WEEK -5	CIRCULAR LINKED LIST
	5. Write a program that uses functions to perform the following operations on circular linkedlist. i) Creation ii) Insertion iii) Deletion iv) Traversal

WEEK -6	SORTING TECHNIQUES – I
6. Write a program that implements the following sorting methods. i) Bubble sort ii) Insertion sort iii) Selection sort	
WEEK -7	SORTING TECHNIQUES - II
7. Write a program that implements the following sorting methods. i) Quick sort ii) Merge sort iii) Heap sort	
WEEK -8	TREE TRAVERSALS
8. Write a program to implement the tree traversal methods. i) Pre-order ii) In order iii) Post order	
WEEK -9	BINARY SEARCH TREES
9. Write a program to implement i) Binary Search tree ii) B Trees	
WEEK -10	BINARY SEARCH TREES
10. Write a program to implement i) B+ Trees ii) AVL trees iii) Red - Black trees	
WEEK -11	GRAPH TRAVERSAL
11. Write a program to implement the graph traversal methods. i) DFS ii) BFS	
WEEK -12	PATTERN MATCHING
12. Implement a Pattern matching algorithms using Brute force, Knuth-Morris-Pratt	
TEXT BOOKS	
1. Fundamentals of Data Structures in C, 2nd Edition, E. Horowitz, S. Sahni and Susan Anderson Freed, Universities Press. 2. Data Structures using C A. S. Tanenbaum, Y. Langsam, and M. J. Augenstein, PHI/Pearson Education.	